UE24CS320A – Capstone Project Approval

A Comparative Framework for Early Sepsis Prediction Using Ensemble and Explainable Deep Learning on ICU Data

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Problem Statement

- Sepsis is the body's extreme reaction to an infection that damages its own organs. Mortality rises every hour treatment is delayed, so **early prediction** 6 to 12 hours before onset is critical. Models already reach **AUROC 0.85–0.91** (ensemble) and about 0.85 (LSTM), with **SHAP** as the default explanation.
- Two gaps remain: ensemble and deep-learning models are **compared on different datasets and protocols**, and **no study checks whether SHAP explanations are stable, faithful and robust** when features are correlated and labs are 85–95% missing.

Current practice

Ensemble	**Deep learning**	**Explanation**
XGBoost, RF	LSTM	SHAP
AUROC 0.85–0.91	AUROC ≈ 0.85	Default method

Predicting well is not enough.

An explanation can change after retraining while the prediction stays the same, and clinicians get no signal that it happened.

Can we compare ensemble and deep learning fairly, and tell when a SHAP explanation can be trusted, fixed or withheld?

Motivation

Global

- About 30 million people develop sepsis each year
- About 6 million die from it each year
- An estimated 4.2 million newborns and children are affected

4–8%

rise in mortality for each hour of delayed treatment

6–12 h

early-warning window before sepsis onset

0.85–0.91

AUROC of ensemble models on PhysioNet 2019

United States

- Nearly 1.7 million develop sepsis and 270,000 die each year
- Over one third of hospital deaths involve sepsis
- Costs hospitals $24 billion a year (13% of healthcare spending)

Major Takeaway: Sepsis is a major problem and early detection saves lives. AI can predict it early, but doctors also need to trust the explanation behind each prediction.

References: Singer et al. (2016); CDC sepsis data reports; WHO sepsis fact sheet; Paoli et al. (2018); Kumar et al. (2006); Seymour et al. (2017); PhysioNet/CinC Challenge 2019

Scope and Feasibility study

Scope

- **Data:** PhysioNet/CinC 2019 (~40,336 ICU patients)
- **Models:** XGBoost, Random Forest, LSTM
- **Explainers:** TreeSHAP, Kernel SHAP, Integrated Gradients
- **Reliability:** stability, faithfulness, robustness
- **Framework:** detect, diagnose, mitigate, abstain
- **Out of scope:** new architectures, AUROC records, prospective clinical validation, multimodal data, real-time deployment

Feasibility

- **Implementation:** the layer wraps existing models; SHAP, Captum, XGBoost and PyTorch are mature.
- **Evaluation:** failure labels are defined before the detector, so evaluation is not circular.
- **Systematic:** perfect reliability is impossible under collinearity, so we detect rather than eliminate.
- **Practical:** a post-hoc layer, not real-time; latency and cost are acceptable.

One Line Conclusion: feasible as a phased prototype on existing models; real-time deployment and prospective validation are future extensions.

References: physionet.org/content/challenge-2019/1.0.0/ | github.com/shap/shap | captum.ai

Applications / Use Cases

1. ICU decision support
A reliability label sits beside every sepsis risk score, so doctors know when not to over-rely on feature attributions.

2. Model auditing
AI governance teams audit deployed sepsis models regularly and catch explanation drift over time.

3. XAI method comparison
Researchers evaluate how reliable different explanation methods are on clinical time-series data.

4. Regulatory support
Quantitative reliability evidence for explainability requirements in AI medical-device submissions.

5. Reusable pattern
Detect, diagnose, mitigate, abstain applies to any clinical prediction task with correlated features and missing data.

Literature Study

Paper / Study	Problem	Approach / Method	Limitations	Relevance to our work
The Attribution Impossibility (2026)	Can feature rankings be faithful, stable, and complete?	Theoretical, not clinical	Formal proof + Lean 4 verification	Mathematical foundation for detecting SHAP instability
First-Mover Bias (2026)	Why does XGBoost SHAP fail?	Isolates mechanism: sequential boosting concentrates on first feature	Not applied to sepsis	Explains failure mode for our XGBoost model
Aas, Jullum & Løland (2021)	How to handle dependent features?	Conditional distribution estimation (Gaussian, Copula)	Not applied to sepsis	Mitigation for feature correlation
LSTM-GAT on PhysioNet 2019, Khorram & Kouchaki (2026)	Early sepsis prediction	LSTM + graph attention, 5 seeds; AUROC 0.853	No reliability evaluation	Deep-learning baseline for comparison
XAI for sepsis systematic review (2026)	How is XAI used in sepsis?	PRISMA review of 37 studies	Diagnoses the gap only	Establishes the need for reliability

Proposed Approach



Blue boxes = our contribution. Output: a reliable explanation, or an "Explanation unreliable" warning.

Reliability score

$$R(x) = w_s \cdot \text{Stability}(x) + w_f \cdot \text{Faithfulness}(x) + w_r \cdot \text{Robustness}(x)$$

Weights start at 1/3 each and are tuned on the validation set.

Example output

"Prediction: 82% sepsis risk. Explanation reliability: LOW. Top feature ranking is unstable across model retraining. Use clinical judgment."

Stack: Python | XGBoost / scikit-learn | PyTorch | SHAP | Captum | SciPy | Pandas / NumPy | Streamlit (optional)

Reliability Layer: Detect

Metric	What we do	Flag it when
A. Stability	Train the same model 10 times with different seeds and compare the explanation for the same patient (Spearman rank correlation, top-k Jaccard overlap)	Jaccard is below 0.6
B. Faithfulness	Get explanations from two methods (SHAP and Integrated Gradients) and see how much their top features overlap	The two methods disagree strongly
C. Robustness	Change one input a little, for example heart rate by ±5 bpm, and see how much the explanation moves (cosine similarity, distance, change rate)	Cosine similarity is below 0.8
D. Temporal stability (LSTM)	Check that the important time steps stay the same at 6 h, 8 h and 10 h before onset	The important time window shifts

Independent failure labels (defined before we build the detector, so the test is fair)

- **Retraining instability**
- **Perturbation sensitivity**
- **Missingness sensitivity**
- **Explainer disagreement**

Diagnose, Mitigate and Abstain

Failure mode	Diagnosis	Mitigation
Feature correlation	Pairwise correlation among top-K features	Dependence-aware attribution (Latent SHAP)
Missingness sensitivity	Compare imputed vs dropped explanations	Informative missingness modelling
Model instability	Retraining variance	Ensemble attribution across models
Perturbation sensitivity	Explanation change under small input changes	ReCalX-style calibration
Temporal instability	Important time window shifts	DeltaSHAP (explains changes between time steps)

Stage 4 – Abstain

If $R(x) < 0.5$, show the risk score with a LOW-reliability warning instead of a feature ranking: “Top feature ranking is unstable across model retraining. Use clinical judgment.”

Novelty Statement

Novelty: the first systematic reliability evaluation framework for SHAP explanations in early sepsis prediction, with detection, diagnosis, mitigation and explanation-level abstention.

Existing work	What is missing
Systematic reviews	Only point out the problem, no solution
Loss landscape research	Uses general data, not sepsis or time-series
LSTM-GAT prediction	No reliability evaluation
ACXAI (2026)	Focuses on generating explanations, not detecting unreliable ones; not sepsis
ReCalX (2025)	Fixes only one failure mode
Attribution Impossibility	Theory only, no clinical framework

Key References

Papers

- Huang & Marques-Silva (2024). On the failings of Shapley values for explainability. IJAR
- Sundararajan & Najmi (2020). The Many Shapley Values for Model Explanation. ICML
- Kumar et al. (2020). Problems with Shapley-value-based explanations. ICML
- Aas et al. (2021). Explaining individual predictions when features are dependent
- Zhao et al. (2024). Error Analysis of Shapley Value-Based Model Explanations
- Hu et al. (2024). Manifold-based Shapley explanations
- UbiQTree (2025). Uncertainty quantification in XAI with tree ensembles
- Khorram & Kouchaki (2026). LSTM-GAT for PhysioNet 2019. BMJ Health & Care Informatics
- MIMIC-Sepsis Benchmark (2025). arXiv:2505.17435

Tools and datasets

- SHAP: github.com/shap/shap
- TimeSHAP: github.com/feedzai/timeshap
- Latent SHAP: github.com/Teriri1999/Latent-SHAP
- UbiQTree: github.com/dubeyakshat07/UBiQTree
- Captum: captum.ai
- PhysioNet 2019: physionet.org/content/challenge-2019/1.0.0/
- MIMIC-IV: physionet.org/content/mimiciv/

Expected Deliverables

Capstone-I – Foundation

- Literature review and novelty matrix
- Dataset preparation and exploratory analysis
- Baselines: XGBoost, RF, LSTM
- Explanation pipeline: SHAP, Integrated Gradients
- Feasibility study and evaluation plan

Milestone: Baselines and explanations running

Capstone-II – Core novelty

- Reliability metrics: stability, faithfulness, robustness
- Independent failure labels defined and generated
- Detector built and evaluated
- Diagnosis and mitigation strategies
- Low-Level Design (LLD)

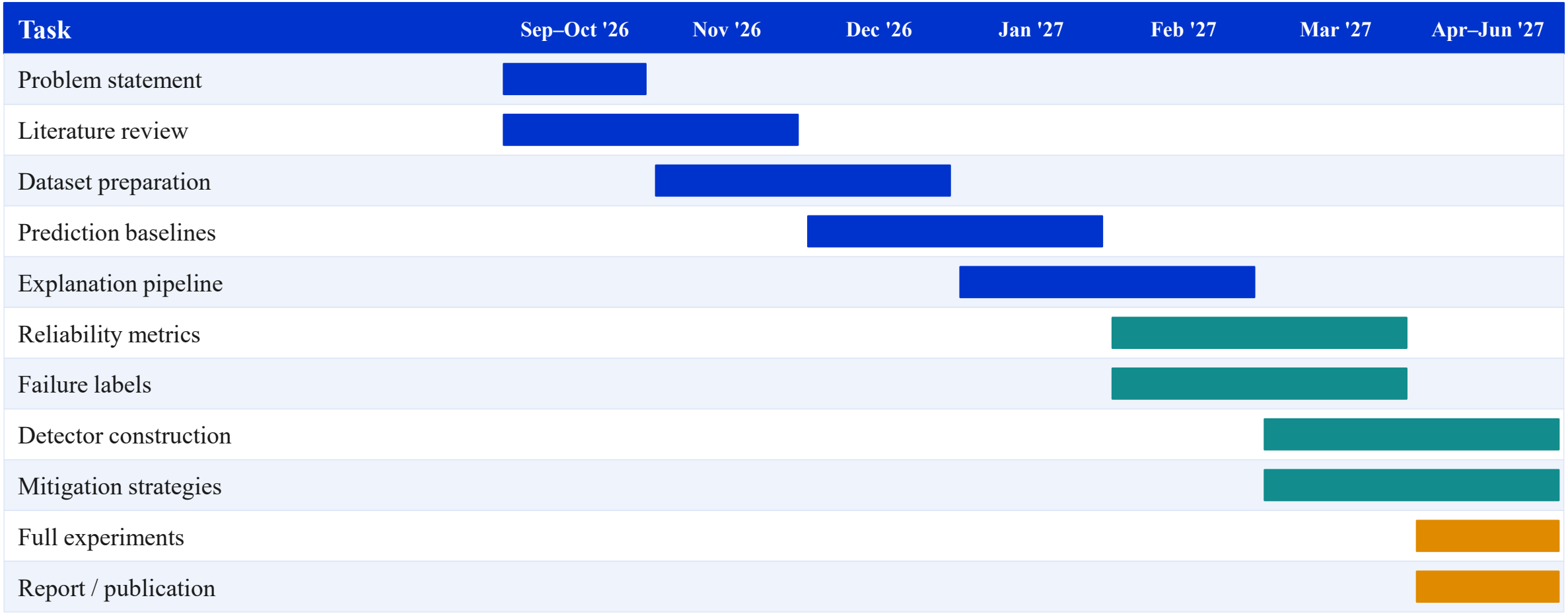
Milestone: Working reliability detector

Capstone-III – Evaluation and paper

- Complete implementation and experiments
- Full evaluation and ablations
- Statistical analysis and visualisation
- Research paper and final documentation
- Reproducibility materials

Milestone: End-to-end framework and paper

Capstone (Phase-I, II & III) Project Timeline



Blue = Capstone-I (foundation) | Teal = Capstone-II (core novelty) | Orange = Capstone-III (evaluation and paper)

Thank You